

Name _____ Period _____ Teacher _____ Test Day _____



Unit 6 – The Mole: Essential Knowledge and Guided Notes

Read Textbook: CK Chemistry Chapter 10.1-10.9: The Mole

<https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

Watch Helpful Videos and Practice: Posted on CANVAS

Self-correcting Practice: <https://tinyurl.com/GeekMolePrac>

THE MOLE

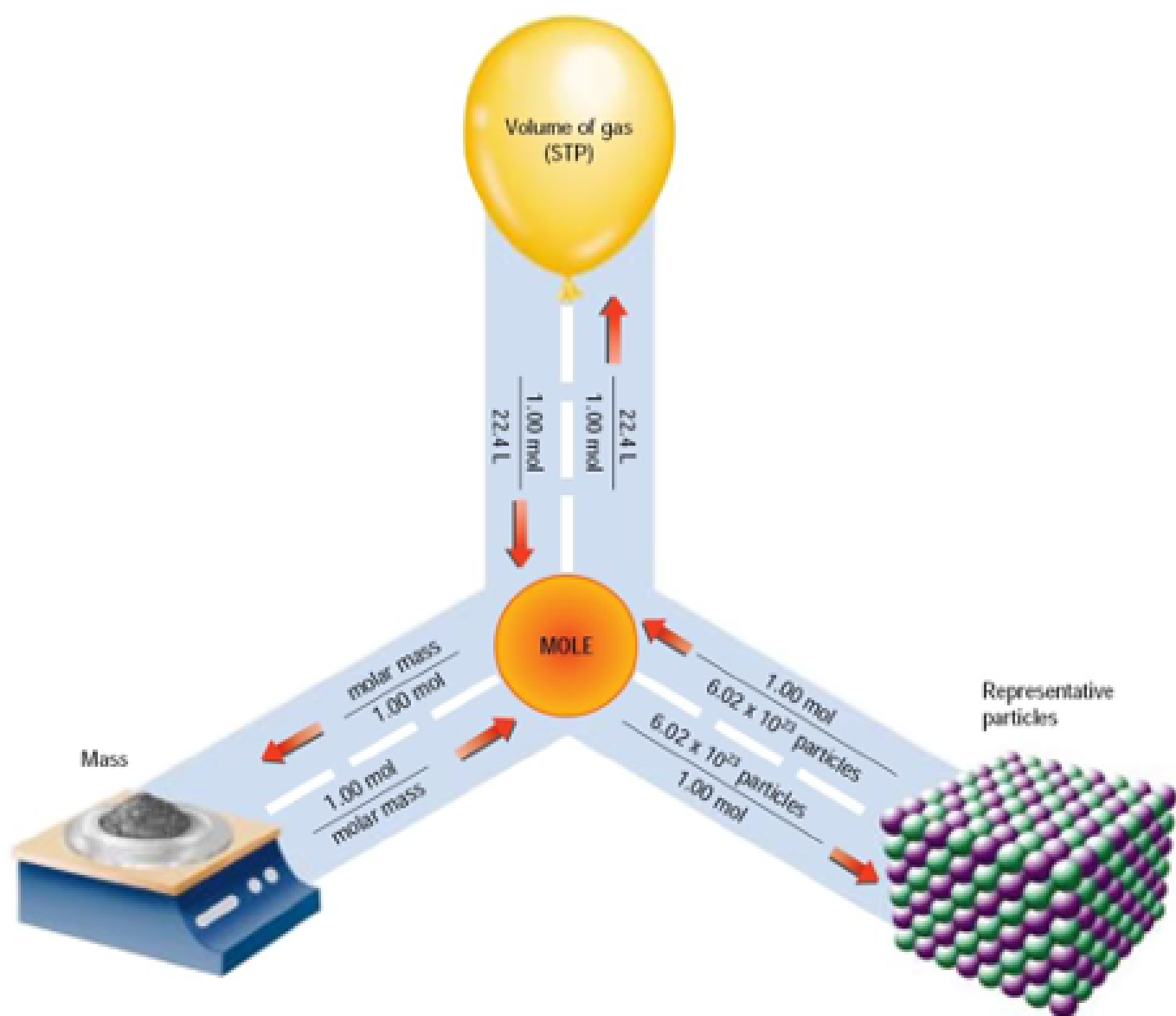
1. Atoms and molecules are too small to count by usual means. A mole is a way of counting any type of particle (atoms, molecules, and formula units).
2. Avogadro's number = 6.02×10^{23} particles per mole.
3. Atomic mass of an ELEMENT (1 mole) is its average atomic mass in grams from the Periodic Table.
4. The Molar (or Formula) Mass of a Compound (1 mole) is the sum of the average atomic masses of all the particles that make it up.

Extra Intensified Content: Some compounds are hydrated with water. Their formulas include the water embedded in the crystal and are written like this: $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$. That is the formula for Copper Sulfate Pentahydrate. You must include the 5 water molecules in the overall molar mass.

$\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$ Molar Mass = $(1 \times 63.55) + (1 \times 32.06) + (4 \times 16.00) + 5 \times (2 \times 1.01 + 1 \times 16.00) = 249.71 \text{ grams}$

5. Molar volume = 22.4 L/mole for any gas at standard temperature and pressure (STP).
6. Standard temperature and pressure is 0°C (273K) and 1 atm (101.3 kPa, 760 mmHg)
7. You can use dimensional analysis to convert between mass, volume, particles, and moles of a substance. The conversion factors used in these calculations are:
 $1 \text{ mole} = 6.02 \times 10^{23}$ atoms or molecules also called representative particles (3 significant figures)
 $1 \text{ mole} = \text{Atomic Mass of the Element or the Molar Mass of the Compound in grams}$
(all masses should be calculated to the hundredths place)
 $1 \text{ mole of a Gas at STP} = 22.4 \text{ L}$ (3 significant figures)

THE MOLE ROAD



Lesson #1: Intro to the Mole

Copy down the examples here:

1. How many atoms of lithium are in 3.50 moles of Lithium?
2. How many moles of lithium are 3.35×10^{24} atoms of lithium?
3. 5.61 moles of Fe = ? atoms Show your work!
4. 9.31×10^{24} molecules of H_2O = ? moles Show your work!
5. How many atoms does your answer for #4 represent?

Lesson #2: The Mole & Mass

Copy down the examples here:

1. Propane is a gas used for cooking. How many molecules are in 2.12 moles of propane (C_3H_8)?
2. How many moles are in 4.65×10^{24} molecules of NO_2 ?
3. How many ATOMS are in 0.75 moles of FeBr_3 ?
4. Find the molar mass of $\text{Al}(\text{NO}_3)_3$

5. Find the molar mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
6. What is the mass of 9.45 moles of Al_2O_3 ?
7. How many moles of Fe_2O_3 are in 92.2 grams?

Lesson #3: Molar Volume & The Mole Road

Copy down the examples here:

1. At STP, how many moles are in 45.7 L of ammonia, NH_3 ?
2. At STP, how many liters are in 9.83 moles of chlorine, Cl_2 ?
3. How many molecules are in 54.0 grams of CO_2 ?
4. How many liters are present in 350 grams of propane (C_3H_8) gas at STP?
5. How many molecules are present in 1200 liters of carbon monoxide gas (CO) at STP?
6. A typical party helium balloon holds about 14.1 L of gas at STP.
 - a) How many atoms of helium are found in such a balloon? (2 steps using mole road)
 - b) What is the mass of helium in the balloon? (2 steps using the mole road)